Sesión de introducción

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Índice

Nueva **1**

Introdución **1**

Software 1

Instalación de paquetes **1**

Nueva

Introdución

Resumen: Para la análisis y modelación de los datos involucrados en la investigación, es importante realizar, como parte del procesamiento de la información, un análisis descriptivo. El cual se compone de gráficos y tablas que muestras las principales características de los datos. El Taller tiene como objetivo usar el Software estadístico R, así como los archivos Markdown y las paqueterías como ggplot2, GGally, janitor, dplyr, forcats, flextable y xtable para generar los reportes en formato Word, PDF y LaTeX.

Software

Enlaces para la descarga del software:

- R-4.6.2 for Windows: https://cran.r-project.org/bin/windows/base/
- Rtools: https://cran.r-project.org/bin/windows/Rtools/rtools45/rtools.html
- RStudio: https://docs.posit.co/ide/user/download.html

Instalación de paquetes

La instalación de paquetes, se realiza con *install.packages()*, si no funciona, se puede usa la función *install_github*, pero para usarla se necesita instalar *library(remotes)* o *library(devtools)*

Otras funciones para la instalación de paquetes es:

a) *install_bioc()*: desde Bioconductor,

- *install_bitbucket()*: desde Bitbucket,
- *install_cran()*: desde CRAN,

1. *install_git()*: desde un repositorio git,

- *install_github()*: desde GitHub,
- *install_local()*: desde un archivo alojado en nuestro equipo,
- *install_svn()*: desde un repositorio SVN,
- *install_url()*: desde una URL,
- *install_version()*: para una versión específica de un paquete de CRAN.

La instalación de los paquetes se hace en dos pasos:

1. Instalación de paquetes: Sólo una vez se instalan.

```
install.packages("flextable")
install.packages("dplyr")

# Paquetes para la compilación del PDF

tinytex::install_tinytex()
tinytex::is_tinytex()
```

2. Activación de los paquetes: Los paquetes siempre se activan.

```
library(flextable)

# library(help=dplyr)

#----
library(dplyr)

library(tinytex) #para compilar PDF
```

La función **dplyr()** ayuda con la manipular las bases de datos:

- *across*: Apply a function (or functions) across multiple columns
- *all_vars*: Apply predicate to all variables
- *arrange*: Order rows using column values
- *auto_copy*: Copy tables to same source, if necessary
- *band_members*: Band membership
- *between*: Detect where values fall in a specified range

- *bind_cols*: Bind multiple data frames by column
- *bind_rows*: Bind multiple data frames by row
- *c_across*: Combine values from multiple columns
- *case-and-replace-when*: A general vectorised if-else
- *coalesce*: Find the first non-missing element
- *compute*: Force computation of a database query
- *consecutive_id*: Generate a unique identifier for consecutive combinations
- *context*: Information about the "current" group or variable
- *copy_to*: Copy a local data frame to a remote src
- *count*: Count the observations in each group
- *cross_join*: Cross join
- *cumall*: Cumulative versions of any, all, and mean
- *desc*: Descending order
- *distinct*: Keep distinct/unique rows
- *dplyr_by*: Per-operation grouping with '.by'/'by'
- *explain*: Explain details of a tbl
- *filter*: Keep or drop rows that match a condition
- *filter-joins*: Filtering joins
- *glimpse*: Get a glimpse of your data
- *group_by*: Group by one or more variables
- *group_cols*: Select grouping variables
- *group_map*: Apply a function to each group
- *group_trim*: Trim grouping structure
- *ident*: Flag a character vector as SQL identifiers
- *if_else*: Vectorised if-else
- *join_by*: Join specifications
- *lead-lag*: Compute lagged or leading values
- *mutate*: Create, modify, and delete columns
- *mutate-joins*: Mutating joins
- *n_distinct*: Count unique combinations
- *na_if*: Convert values to 'NA'
- *near*: Compare two numeric vectors
- *nest_join*: Nest join

- *nth*: Extract the first, last, or nth value from a vector
- *ntile*: Bucket a numeric vector into 'n' groups
- *order_by*: A helper function for ordering window function output
- *percent_rank*: Proportional ranking functions
- *pick*: Select a subset of columns
- *pull*: Extract a single column
- *recode*: Recode values recode-and-replace-values, Recode and replace values
- *reframe*: Transform each group to an arbitrary number of rows
- *relocate*: Change column order
- *rename*: Rename columns
- *row_number*: Integer ranking functions
- *rows*: Manipulate individual rows
- *rowwise*: Group input by rows
- *scoped*: Operate on a selection of variables
- *select*: Keep or drop columns using their names andtypes
- *setops*: Set operations
- *slice*: Subset rows using their positions
- *sql*: SQL escaping.
- *starwars*: Starwars characters
- *storms*: Storm tracks data
- *summarise*: Summarise each group down to one row
- *tbl*: Create a table from a data source
- *vars*: Select variables
- *when-any-all*: Elementwise 'any()' and 'all()'

La función **flextable()** ayuda con la generación de tablas:

- *add_body*: Add body rows with one value per column
- *add_body_row*: Add a body row with spanning labels
- *add_footer*: Add footer rows with one value per column
- *add_footer_lines*: Add full-width rows to the footer
- *add_footer_row*: Add a footer row with spanning labels
- *add_header*: Add header rows with one value per column
- *add_header_lines*: Add full-width rows to the header

- *add_header_row*: Add a header row with spanning labels
- *align*: Set text alignment
- *append_chunks*: Append chunks to flextable content
- *as_b*: Bold chunk
- *as_bracket*: Bracket chunk
- *as_chunk*: Text chunk
- *as_equation*: Equation chunk
- *as_flextable*: Method to transform objects into flextables
- *as_flextable.TableTree*: Transform an rtables object into a flextable
- *as_flextable.compact_summary*: Transform a 'compact_summary' object into a flextable
- *as_flextable.data.frame*: Transform and summarise a 'data.frame' into a flextable Simple summary of a data.frame as a flextable
- *as_flextable.gam*: Transform a 'gam' model into a flextable
- *as_flextable.glm*: Transform a 'glm' object into a flextable
- *as_flextable.grouped_data*: Transform a 'grouped_data' object into a flextable
- *as_flextable.htest*: Transform a 'htest' object into a flextable
- *as_flextable.kmeans*: Transform a 'kmeans' object into a flextable
- *as_flextable.lm*: Transform a 'lm' object into a flextable
- *as_flextable.merMod*: Transform a 'merMod' or 'lme' object into a flextable
- *as_flextable.pam*: Transform a 'pam' object into a flextable
- *as_flextable.summarizor*: Transform a 'summarizor' object into a flextable
- *as_flextable.table*: Transform a 'table' object into a flextable
- *as_flextable.tabular*: Transform a 'tables::tabular' object into a flextable
- *as_flextable.tabulator*: Transform a 'tabulator' object into a flextable
- *as_flextable.xtable*: Transform a 'xtable' object into a flextable
- *as_grouped_data*: Insert group-label rows into a data frame
- *as_highlight*: Highlight chunk
- *as_i*: Italic chunk
- *as_image*: Image chunk
- *as_paragraph*: Build a paragraph from chunks
- *as_qmd*: Quarto inline markdown chunk
- *as_strike*: Strikethrough chunk
- *as_sub*: Subscript chunk

- *as_sup*: Superscript chunk
- *as_word_field*: Word dynamic field chunk
- *autofit*: Adjust columns to their content size
- *before*: Detect rows before a given value
- *bg*: Set background color
- *body_add_flextable*: Add flextable into a Word document
- *body_replace_flextable_at_bkm*: Add flextable at bookmark location in a Word document
- *bold*: Set bold font
- *border_inner*: Set all inner borders
- *border_inner_h*: Set inner horizontal borders
- *border_inner_v*: Set inner vertical borders
- *border_outer*: Set outer borders
- *border_remove*: Remove borders
- *colformat_char*: Format character cells
- *colformat_date*: Format date cells
- *colformat_datetime*: Format datetime cells
- *colformat_double*: Format double cells
- *colformat_image*: Format cells as images
- *colformat_int*: Format integer cells
- *colformat_lgl*: Format logical cells
- *colformat_num*: Format numeric cells with format()
- *color*: Set font color
- *colorize*: Colorize chunk
- *compact_summary*: Compact Summary of a Dataset
- *compose*: Set cell content from paragraph chunks
- *continuous_summary*: Summarize continuous variables as a flextable
- *cross_join*: Cross join (Nota: Esta función pertenece a paquetes como dplyr)
- *delete_columns*: Delete flextable columns
- *delete_part*: Delete flextable part
- *delete_rows*: Delete flextable rows
- *df_printer*: data.frame automatic printing as a flextable
- *dim.flextable*: Get column widths and row heights of a flextable
- *dim.flextableGrob*: Get optimal width and height of a flextable grob

- *dim_pretty*: Calculate optimal column widths and row heights
- *empty_blanks*: Make blank columns transparent
- *fit_columns*: Constrain table width by wrapping text
- *fit_to_width*: Constrain table width by shrinking font size
- *flextable*: Create a flextable from a data frame
- *flextable-package*: flextable: Functions for Tabular Reporting
- *flextable_dim*: Get overall width and height of a flextable
- *flextable_to_rmd*: Print a flextable inside knitr loops and conditionals
- *fmt_2stats*: Format summarizor statistics as text
- *fmt_avg_dev*: Format mean and standard deviation as text
- *fmt_dbl*: Format numbers as doubles
- *fmt_header_n*: Format count as '(N=XX)' for column headers
- *fmt_int*: Format numbers as integers
- *fmt_n_percent*: Format count and percentage as text
- *fmt_pct*: Format numbers as percentages
- *fmt_signif_after_zeros*: Format with significant figures after zeros
- *font*: Set font
- *fontsize*: Set font size
- *footnote*: Add footnotes to flextable
- *fp_border_default*: Create border formatting with flextable defaults
- *fp_text_default*: Create text formatting with flextable defaults
- *gen_grob*: Render a flextable as a graphic object
- *get_flextable_defaults*: Get flextable defaults formatting properties
- *gg_chunk*: ggplot chunk
- *grid_chunk*: Grid Graphics chunk
- *height*: Set flextable rows height
- *highlight*: Set text highlight color
- *hline*: Set horizontal borders below selected rows
- *hline_bottom*: Set the bottom border of a table part
- *hline_top*: Set the top border of a table part
- *hrule*: Set how row heights are determined
- *htmltools_value*: Convert a flextable to an HTML object
- *hyperlink_text*: Hyperlink chunk

- *indentation*: Set paragraph indents
- *italic*: Set italic font
- *keep_with_next*: Set Word 'Keep with next' instructions
- *knit_print.flextable*: Render flextable in knitr documents
- *labelizor*: Replace displayed text with labels
- *line_spacing*: Set line spacing
- *linerange*: Mini linerange chunk
- *merge_at*: Merge flextable cells into a single one
- *merge_h*: Merge flextable cells horizontally
- *merge_h_range*: Rowwise merge of a range of columns
- *merge_none*: Delete flextable merging information
- *merge_v*: Merge flextable cells vertically
- *minibar*: Mini barplot chunk
- *ncol_keys*: Number of columns
- *nrow_part*: Number of rows of a part
- *padding*: Set paragraph paddings
- *paginate*: Prevent page breaks inside a flextable
- *ph_with.flextable*: Add a flextable into a PowerPoint slide
- *plot.flextable*: Plot a flextable
- *plot.flextableGrob*: plot a flextable grob
- *plot_chunk*: Mini plot chunk
- *prepend_chunks*: Prepend chunks to flextable content
- *print.flextable*: Print a flextable
- *proc_freq*: Frequency table
- *repair_docx*: Repair flextables in a 'Word' document produced by 'pandoc'
- *rotate*: Cell text rotation
- *rtf_add.flextable*: Add a 'flextable' into an RTF document
- *save_as_docx*: Save flextable objects in a 'Word' file
- *save_as_html*: Save flextable objects in an 'HTML' file
- *save_as_image*: Save a flextable in a 'png' or 'svg' file
- *save_as_pptx*: Save flextable objects in a 'PowerPoint' file
- *save_as_rtf*: Save flextable objects in an 'RTF' file
- *save_as_typst*: Save flextable objects in a Typst file

- *separate_header*: Split column names using a separator into multiple rows
- *set_caption*: Set flextable caption
- *set_flextable_defaults*: Modify flextable defaults formatting properties
- *set_formatter*: Set column formatter functions
- *set_header_footer_df*: Replace the entire header or footer from a data frame
- *set_header_labels*: Rename column labels in the header
- *set_table_properties*: Set table layout and width properties
- *shift_table*: Create a shift table
- *split_columns*: Split a flextable into pages by columns
- *split_rows*: Split a flextable into pages by rows
- *split_to_pages*: Split a flextable into pages by rows and columns
- *style*: Set formatting properties on a flextable selection
- *summarizor*: Prepare descriptive statistics for flextable
- *surround*: Surround cells with borders
- *tab_settings*: Set tabulation marks configuration
- *tabulator*: Create pivot-style summary tables
- *tabulator_colnames*: Column keys of tabulator objects
- *theme_alafoli*: Apply alafoli theme
- *theme_apa*: Apply APA theme
- *theme_booktabs*: Apply booktabs theme
- *theme_borderless*: Apply borderless theme
- *theme_box*: Apply box theme
- *theme_tron*: Apply tron theme
- *theme_tron_legacy*: Apply tron legacy theme
- *theme_vader*: Apply Sith Lord Darth Vader theme
- *theme_vanilla*: Apply vanilla theme
- *theme_zebra*: Apply zebra theme
- *to_html.flextable*: Get HTML code as a string
- *use_df_printer*: Set data.frame automatic printing as a flextable
- *use_flextable_qmd*: Install the flextable-qmd Quarto extension
- *use_model_printer*: Set automatic flextable printing for models
- *valign*: Set vertical alignment
- *vline*: Set vertical borders to the right of selected columns

- *vline_left*: Set the left border of the table
- *vline_right*: Set the right border of the table
- *void*: Clear the displayed content of selected columns
- *width*: Set columns width
- *wrap_flextable*: Wrap a flextable for use with patchwork